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ROLL NO – MC2550 (MCA)

Assignment 1

Find 5 AI systems you used today. Classify each as ML, DL, NLP, or GenAI. Explain what data they likely learned from and how they affect your daily life.

Answer:

Today, I used several AI-based systems in my daily activities. Five of them are explained below along with their classification, the type of data they likely learned from, and their impact on my daily life.

1. ChatGPT

Classification: Generative AI (GenAI)

What data did it likely learn from?

ChatGPT was trained using large amounts of text data, including publicly available information, licensed data, and data created by human trainers. This helps the model learn patterns in language and generate relevant responses.

How does it affect my daily life?

I use ChatGPT to understand difficult concepts, clear doubts, get help with assignments, generate ideas and learn new topics. It makes studying and finding explanations faster and easier.

2. Google Search

Classification: Machine Learning (ML)

What data did it likely learn from?

Google Search uses information from web pages and signals related to searches and content to learn patterns that help rank and retrieve relevant information.

How does it affect my daily life?

I use Google Search to find study materials, websites, tutorials and information about different topics. It saves time by showing relevant results for my queries.

3. YouTube Recommendation System

Classification: Deep Learning (DL)

What data did it likely learn from?

The recommendation system can use signals such as videos watched, search history, likes, subscriptions, watch time and interactions to identify patterns and recommend content.

How does it affect my daily life?

YouTube recommends educational and entertainment videos according to my interests. This helps me discover useful content without manually searching for every video.

4. Google Translate

Classification: Natural Language Processing (NLP)

What data did it likely learn from?

Google Translate uses large amounts of multilingual text and examples across different languages to learn relationships between languages and how sentences can be translated.

How does it affect my daily life?

I use Google Translate to understand unfamiliar words and translate sentences between different languages. It makes communication and understanding content in other languages easier.

5. Gmail Spam Detection

Classification: Machine Learning (ML)

What data did it likely learn from?

Spam-detection systems learn from patterns and signals found in large numbers of legitimate and spam emails. These may include message content, sender information, links and other characteristics associated with unwanted messages.

How does it affect my daily life?

Gmail automatically identifies suspicious or unwanted emails and places them in the spam folder. This keeps my inbox organised and reduces exposure to potentially harmful messages.

Assignment 2

Visit Google Colab. Paste 3 sentences. Record token counts. Try a sentence in another language. What patterns do you observe in how words are split?

Sentence	Token Count
Artificial Intelligence is transforming education.	7
I enjoy learning about Large Language Models and Generative AI.	12
ChatGPT helps students understand complex concepts easily.	10
आज मौसम अच्छा है। (HINDI)	21
आज हवामान छान आहे. (MARATHI)	20
આજે હવામાન સારું છે. (GUJRATI)	33

From the tokenizer experiment, I observed the following patterns:

1. **One word does not always equal one token.** Some words are represented as a single token, while longer or less common words may be divided into multiple tokens.
2. **Common words are often represented using fewer tokens,** whereas uncommon or complex words may be split into smaller parts.
3. **Spaces and punctuation also affect tokenization.** Punctuation marks such as ., ,, ?, and ! may be separate tokens or combined with nearby text.
4. **Different languages are tokenized differently.** Hindi, Marathi, and Gujarati sentences may require more tokens than similar English sentences because the tokenizer represents different scripts and character sequences differently.

5. **Token count is not the same as word count.** A sentence containing only a few words can still contain many tokens.

```
!pip install tiktoken
import tiktoken

encoding = tiktoken.get_encoding("cl100k_base")

print("Tokenizer loaded successfully!")
import tiktoken

# Load tokenizer
encoding = tiktoken.get_encoding("cl100k_base")

# Sentences
sentences = {
    "English 1": "Artificial Intelligence is transforming education.",
    "English 2": "I enjoy learning about Large Language Models and Generative AI.",
    "English 3": "ChatGPT helps students understand complex concepts easily.",
    "Hindi": "आज मौसम अच्छा है।",
    "Marathi": "आज हवामान छान आहे.",
    "Gujarati": "આજે હવામાન સારું છે."
}

# Tokenize all sentences
for language, sentence in sentences.items():

    tokens = encoding.encode(sentence)

    print("\nLanguage:", language)
    print("Sentence:", sentence)
    print("Tokens:", tokens)
    print("Token Count:", len(tokens))

    print("Token Breakdown:")

    for token in tokens:
        print(repr(encoding.decode([token])))

    print("-" * 50)
```

```
Requirement already satisfied: tiktoken in /usr/local/lib/python3.12/dist-packages (0.13.0)
Requirement already satisfied: regex in /usr/local/lib/python3.12/dist-packages (from tiktoken) (2025.11.3)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from tiktoken) (2.32.4)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->tiktoken) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->tiktoken) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->tiktoken) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->tiktoken) (2026.6.17)
Tokenizer loaded successfully!
```

```
Language: English 1
Sentence: Artificial Intelligence is transforming education.
Tokens: [9470, 16895, 22107, 374, 46890, 6873, 13]
Token Count: 7
Token Breakdown:
'Art'
'ificial'
' Intelligence'
' is'
' transforming'
' education'
'.'
```

```
Language: English 2
Sentence: I enjoy learning about Large Language Models and Generative AI.
Tokens: [40, 4774, 6975, 922, 20902, 11688, 27972, 323, 2672, 1413, 15592, 13]
Token Count: 12
Token Breakdown:
'I'
' enjoy'
' learning'
' about'
' Large'
' Language'
' Models'
' and'
' Gener'
' ative'
' AI'
'.'
```

```
Language: English 3
Sentence: ChatGPT helps students understand complex concepts easily.
Tokens: [16047, 38, 2898, 8779, 4236, 3619, 6485, 19476, 6847, 13]
Token Count: 10
Token Breakdown:
'Chat'
'G'
'PT'
' helps'
' students'
' understand'
' complex'
' concepts'
' easily'
','
```

```

Language: Hindi
Sentence: आज मौसम अच्छा है।
Tokens: [5619, 228, 5619, 250, 92317, 12906, 234, 79468, 88344, 15272, 227, 5619, 248, 31584, 249, 24810, 85410, 12906, 230, 12906, 97]
Token Count: 21
Token Breakdown:
'।'
'है'
'।'
'है'
'।'
'है'
'।'
'म'
'।'
'स'
'।'
'म'
'।'
'स'
'।'
'म'
'।'
'स'
'।'
'म'

```

```
Language: Marathi
Sentence: आज हवामान छान आहे.
Tokens: [5619, 228, 5619, 250, 85410, 5619, 113, 32511, 106, 32511, 101, 15272, 249, 32511, 101, 15272, 228, 95048, 35470, 13]
Token Count: 20
Token Breakdown:
'।'
'।'
'।'
'।'
'।'
'ह'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
'।'
```

```

Language: Gujarati
Sentence: અજિ હવમામ સારં છે.
Tokens: [41814, 228, 41814, 250, 73319, 229, 95591, 117, 41814, 113, 41814, 122, 41814, 106, 41814, 122, 41814, 101, 95591, 116, 41814, 122]
Token Count: 33
Token Breakdown:
[{"token": "41814", "count": 10, "order": 1}, {"token": "228", "count": 1, "order": 2}, {"token": "41814", "count": 1, "order": 3}, {"token": "250", "count": 1, "order": 4}, {"token": "73319", "count": 1, "order": 5}, {"token": "229", "count": 1, "order": 6}, {"token": "95591", "count": 2, "order": 7}, {"token": "117", "count": 1, "order": 8}, {"token": "41814", "count": 1, "order": 9}, {"token": "113", "count": 1, "order": 10}, {"token": "41814", "count": 1, "order": 11}, {"token": "122", "count": 2, "order": 12}, {"token": "41814", "count": 1, "order": 13}, {"token": "106", "count": 1, "order": 14}, {"token": "41814", "count": 1, "order": 15}, {"token": "122", "count": 1, "order": 16}, {"token": "41814", "count": 1, "order": 17}, {"token": "101", "count": 1, "order": 18}, {"token": "95591", "count": 1, "order": 19}, {"token": "116", "count": 1, "order": 20}, {"token": "41814", "count": 1, "order": 21}, {"token": "122", "count": 1, "order": 22}
]

```